Faculty of Engineering and Technology

Electrical and Computer Engineering Department

MACHINE LEARNING AND DATA SCIENCE-ENCS5341

Report

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1. Introduction

This project focuses on classifying weather conditions from images using machine learning techniques. A frozen pre-trained Convolutional Neural Network (CNN) was used to extract high-level feature representations from images, which were stored and reused for efficient training and evaluation. Three models were explored: K-Nearest Neighbors (KNN) as a baseline, Random Forest as an ensemble-based approach, and a non-linear Support Vector Machine (SVM). Model performance was evaluated using accuracy, precision, recall, and F1-score to compare the effectiveness of classical machine learning models on CNN-extracted features.

Table 1:Metrics

Model	Accuracy	F1	Note
KNN	0.7136	0.70	K =1 (Best model)
RANDOM FOREST	0.704225352112676	0.61	Tuned
SVM	0.6995	0.67	Best model

Best Model = KNN (K =1)

2. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the dataset structure and identify potential challenges. Data cleaning steps included removing duplicate entries, correcting spelling inconsistencies, handling missing values, and removing irrelevant attributes. Some issues were fixed programmatically, while others required manual correction. A significant class imbalance was observed, with the 'Sunny' weather class dominating the dataset.

Descriptive analysis revealed that weather and season categories were unevenly distributed, which may bias model learning. Visual inspection showed strong correlations between certain weather conditions and mood labels, as well as overlapping visual characteristics between classes such as Sunny and Cloudy. Due to the focus on weather classification, visualizations were limited to Weather and Season distributions.

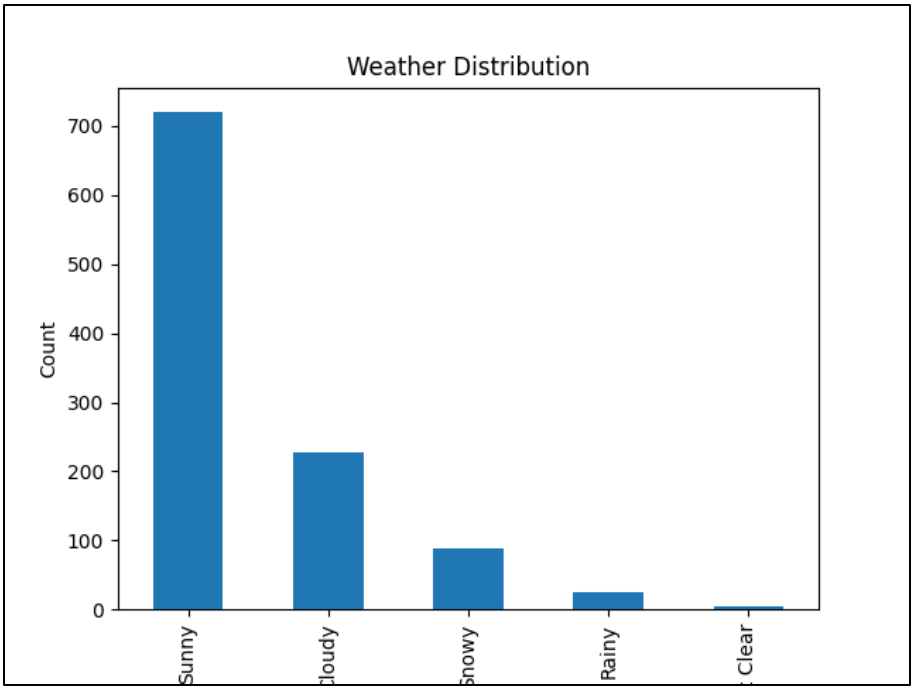

Figure 1:Weather Visualization

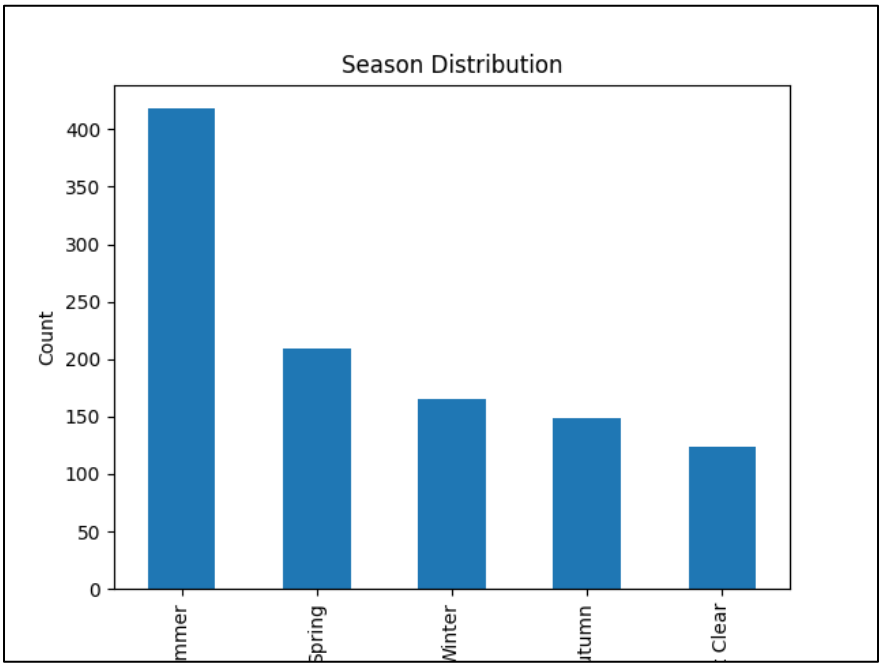

Figure 2:Season Visualization

Note: Visualizations were created only for the *Season* and *Weather* variables because the primary focus of this work was on weather-based image analysis.

3. Experiments and Results

All experiments were conducted using CNN-extracted feature vectors stored in a compressed .npz file. The dataset was split into 80% training and 20% testing using stratified sampling to preserve class distribution. Performance was evaluated using accuracy, confusion matrices, and classification reports.

The KNN baseline model was evaluated with k=1 and k=3, providing a simple distance-based reference. Random Forest was trained with 300 trees and further tuned using grid search, while the SVM model employed an RBF kernel with standardized features. The Random Forest achieved an accuracy of approximately 0.704, slightly outperforming SVM at 0.6995, while KNN achieved comparable performance.

Hyperparameter tuning for Random Forest did not yield improvement, indicating that performance limitations stem from dataset characteristics rather than model configuration. Overall, results across all models were comparable.

4. Analysis

Error analysis was conducted by examining misclassified test samples across all models. Misclassifications were concentrated in minority classes such as Rainy and Not Clear, which were frequently predicted as the dominant Sunny class. Confusion matrices revealed consistent confusion between visually similar categories, particularly Sunny and Cloudy.

High-confidence incorrect predictions, especially in SVM and Random Forest models, suggest the presence of ambiguous or mislabeled samples. The KNN baseline further demonstrated sensitivity to local neighborhood bias caused by class imbalance. These findings highlight that dataset quality and class distribution significantly influence model performance.

5. Conclusions and Discussion

This study demonstrated that classical machine learning models can achieve reasonable performance when applied to CNN-extracted image features. Random Forest slightly outperformed SVM and KNN; however, no model showed substantial improvement after tuning.

Key limitations include strong class imbalance, limited data for minority classes, and visual similarity between weather categories. Future work should focus on collecting more balanced data, applying advanced data augmentation, and exploring fine-tuning of CNN models or transformer-based architectures for improved feature representation.